MIL-T-48419A(MU)
1 December 1975
SUPERSEDING
MIL-T-48419(MU)
25 April 1974

MILITARY SPECIFICATION

TESTER, SYSTEM, : 10549290

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for the System Tester, 10549290.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling, Procedure and Tables for Inspection by Attributes
M1L-STD-109	Quality Assurance Terms and Definitions
MIL-STD-454	Standards, General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

DRAWING

Military

10549290 Tester, System

PACKAGING DATA SHEET

10549290 Packaging of Tester, System

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Identification. - The Tester consists of electrical and mechanical devices and indicators which are used as cursory checks of the proficiency of the XM1O System in lieu of the Vulcan gun.

3.2 Characteristics.

3.2.1 Performance. - The Tester shall be manufactured as specified in the applicable specification and drawing 10549290.

3.2.1.1.1 Temperature. - The Tester shall be capable of being stored at +160°F and -67°F, and operated after being temperature stabilized at +140°F and -4°F.Subsequent to such exposure the Tester shall not have been damaged or adversely affected and shall meet electrical Performance requirements of this specification at standard room ambient conditions.

3.2.1.1.2 Humidity.

3.2.1.1.2.1 Operating. - The Tester shall be capable of meeting the performance requirements of this specification while exposed to an atmosphere with a relative humidity (RH) of 94 ± 4 percent at a temperature of 86°F.

3.2.1.1.2.2 Non-operating. - The Tester shall be capable of withstanding 240 hours of continuous exposure to a relative humidity (RH) of 94 ± 4 percent at temperature cycling of from 73°F to 140°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

3.2.1.1.3 Shock. - The Tester shall be capable of withstanding three 10G, 11 milliseconds in each of its mutually perpendicular axis for a total of 18 shocks.

3.2.1.1.4 Vibration. - The Tester shall be capable of withstanding a vibration having a double amplitude of 0.076 in. and acceleration levels of from 2 to 4G applied logarithmically over the 5 to 500 Hz frequency range for a 15 minute time duration.

3.2.1.1.5 Salt fog. The Tester shall be capable of withstanding forty-eight (48) hours of continuous exposure to a 5% salt solution with a specific gravity of 1.027 to 1.034, a PH of 6.5 to 7.2 and a temperature of 95°F.

after three (3) hours of exposure to wind driven sand and dust.

3.2.1.1.7 Rain. - The Tester shall be capable of operation after exposure to wind driven rain.

3.2.1.1.8 Altitufde. - The Tester shall be capable of operating during exposure at an altitude of 10,000 ft. and after exposure at an altitude of 50,000 ft.

3.2.1.1.9 Fungus. - The Tester shall be capable of withstanding a twenty-eight (28) day fungus exposure to a composite spore suspension.

3.3 Design and Construction.

3.3.1 Production Drawings. - The Tester shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents listed on the applicable specification and assembly drawing,
Any conflict between this specification and the drawing, the drawing shall be considered a superseding requirement.

3.3.2 Standard of Manufacturing. The Tester shall be manufactured to meet the requirements of applicable drawing 10549290 , MIL-F-13926 and MIL-C-46357.

3.3.2.1 MIL-F-13926.

a. Order of-precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishing
d. Part identification and marking
e. Soldering electrical connections

3.3.2.2 MIL-C-46357.

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

3.4 Preproduction Sample. - Preproduction samples shall be given qualification test prior regular production to demonstrate the adequacy and suitability of the contractor's processes and procedures in achieving the performance that is inherent in the design. (See 6.2)

3.5 Electrical.

3.5.1 Continuity. -The Tester shall demonstrate continuity in accordance with all the wiring requirements as cited on the drawings/ wiring list constituting part of the Technical Data Package.

3.5.2 The Tester shall be capable of providing VADS simulated signals when the Tester is connected into XM10 system.

3.5.2.1 The Tester shall be capable of applying the voltages of Table I to the connector pins of Table I as selected by S1.

TABLE I

SWITCH POSITION	P2 CONNECTOR PINS	DC VOLTAGE
1	B	+28V
2	V	+28V
3	W	+28V
4	W	+28V
	X	+13.34V
5	b	+28V
	X	+13.34V
6	e	+28V
	v	+28V
7	e	+28V
	v	+28V
	q	+28V
8	w	+28V
	v	+28V
9	x	+28V
10	$\overline{y}$	+28V
	H	+5.0V
11	Y	+28V
12	V	+28V
	$\overline{a}$	+2.33V

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

SWITCH POSITION	P2 CONNECTOR PINS	DC VOLTAGE
14	X	+28V
	v	+28V
15	v	+28V
	c	+14V
16	f	+28V
17	h	+28V
18	r	+3.65V
	f	+28V
19	EE	+22.58V
20	Z	+12.05V
21	P	-3.73V
	n	+2V

3.5.3 The Tester shall be capable of a meter indication of signals originating in the XM10 that are received by the VADS. The signals that are monitored by the Tester are +28V, +5V, + and -6, + and -10, + and -15, and +20. (All voltage DC)

3.5.4 The Tester shall provide indicator lamp to indicate applications of +28V, -28V, and 115V ac.

3.5.5 The Tester shall provide a synchro transmitter output.

3.5.6 The Tester shall provide for audiocommunication between Tester and XM10 system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISION

4.1 General.

4.1.1 Responsibility for inspection. - Unless otherwise specified in the contact, the supplier is responsible for the performance of all inspection requirements as specified herein. Wxcept as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved-by the-Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections-are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1.1 General provisions. - The componemet and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.1.2 Special tests and examinations.

4.1.2.1 First article (initial production) approval. - The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall be 2 assemblies manufactured in the same manner using the same materials, equipment, processes, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The assemblies shall meet the requirements and tests of Table II.

4.1.2.1.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this insepction with the sample, including certificates of conformance for materials.

4.1.2.1.2 Contractor testing. - When the contractor is responsible for conduction first article aproval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and record of this inspection including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.1.2.2 Control tests. - Three assemblies shall be selected at random as a control sample from each 50 produced. The control samples shall meet the requirements and tests of Table II. (See 6.3)

4.1.2.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (See 4.2.2),

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

I

TABLE II

TEST AND REQUIREMENTS

TEST	REQUIREMENT	TEST PROCEDURES
Performance	3.6	4.3.12
Temperature	3.2.1.1.1	4.3.1
Humidity	3.2.1.1.2	4.3.2
Shock	3.2.1.1.3	4.3.3
Vibration	3.2.1.1.4	4.3.4
Salt fog	3.2.1.1.5	4.3.5
Sand and Dust	3.2,1.1.6	4.3.6
Rain	3.2.1.1.7	4.3.7
Altitude	3.2.1.1.8	4.3.8
Fungus	3.2.1.1.9	4.3.9
Design and Construction	3.3	Visual

4.2 Quality conformance inspections.

4.2.1 Submission of product. - Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots-shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.3 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.2.3.1 Contractor designed equipment. - The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in the detail specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerances limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

4.2.3.2 Test equipment. - Test equipment used to test assemblies tor verification of Section 3 electrical requirements shall be as specified in paragraph 4.3.10.

4.2.4 Examinations. - Unless otherwise, specified, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations to verify requirements of 3.3 shall be performed. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.5 Fuctional tests. - Detail functional requirements specified in the detail specification shall be inspected on a 100 percent basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature	+50°F to +90°F
b. Relative humidity	90 percent RH max.
c. Pressure	28 to 32 inches of Hg

4.3 Environmental Test Methods and Procedures.

4.3.1 Temperature Storage.

a. High Temperature. - This test shall be conducted in accordance with Procedure I, Method 501 of MIL-STD-810B.

b. Low Temperature. - This test shall be conducted in accordance with Procedure I, Method 502, of MIL-STD-81OB.

4.3.2 Humidity. - The Tester shall be subjected to the humidity test as outlined in MIL-STD-810B, Method 507, Procedure II, except that performance test to be conducted during the exposure shall be conducted during the 1st, 3rd, and 5th cycles; subsequently the Tester shall not be damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.3 Shock. - The Tester shall be subjected to three 10G, 11 millisecond duration, shocks in each of the three mutually perpendicular directions for a total of 18 shocks as outlined in MIL-STD-810B, Procedure I. Subsequent to the shock test, the Tester shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.4 Vibration. - The Tester shall be subjected to the vibration as specified in MIL-STD-810, Method 514, Procedure VIII except that the displacement and frequency shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

GRAVITY	FREQUENCY (Hz)
0.076 in double amplitude	5-23
2 G	23-40
4 G	40-500

Subsequent to the vibration test, the Tester shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.5 Salt fog. - The Tester shall be subjected to the salt fog test as specified in MIL-STD-810, Method 509. Subsequent to the exposure the Tester shall not be damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.6 Sand and Duct. - The Tester shall be subjected to the sand and dust test as specified in MIL-STD-810, Method 510. Subsequent to the exposure the Tester shall not be damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.7 Rain. - The Tester shall be subjected to the rain test as specified in MIL-STD-810, Method 506. The tester will not be operated during the test, but subsequently it shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.8 Altitude. - The Tester shall be subjected to the altitude as specified in MIL-STD-81O, Method 500, Procedure I. Subsequent to the exposure, the Tester shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.3.9 Fungus. - The Tester shall be subjected to the Fungus Test as specified in MIL-STD-810, Method 508, for 28 days. Subsequent to the exposure the Tester shall not have been damage or adversely affected and shall meet the performance requirements of this specification.

4.3.10 Test Equipment.

ITEM	DESCRIPTION	
Power Supply	Range	0 - 50 VDc 115 VAC 400 Hz
	Accuracy	± 1.0%
	Sensitivity	20K OHM/VDC
Voltmeter, Digital	Range	0 - 500 VDC 0 - 250 VAC
	Accuracy and Resolution	0.1% or better across the range.
	Input Impedance	- 10 MEG or better for all voltage scales.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

Oscilloscope	Rise Time for scope and associated measuring probe of 50 nanoseconds or greater, Sweep period - 50 nanoseconds per centimeter or greater. Input Resistance of greater than 8 megohms. Input capacitance of less than 25 Picafarads.
Synchro	As shown on Part No. 11740911

4.3.11 Electrical

4.3.11.1 Continuity. - The Tester shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.5.1.

4.3.12.1 With +28 ± 0.1V applied to J10-C, -28 ± 0.1V to J10-X, +15 + 0.1V to J10-K, and ground to J10-G, the voltage outputs of-connector P2 shall be as listed in Table I. The tolerances of the voltages at P2 connector shall be ± 5%. After completion of test remove all connections from System.

4.3.12.2 Apply +28V ± 0.05V to J10-C and ground to P2-u. Place S2 to position 1, S1 to position 13, meter M1 shall read 50 ± 5. Repeat the above procedure by applying the appropriate voltage and ground connection with its associated S2 switch position as listed in Table III. Meter M1 shall read 50 ± 5 in position 1 through 12, inclusive. The tolerance of V input voltages shall be ± 0.05V. After completion of test remove all connections from System Tester.

TABLE III

S2 SWITCH POSITION	V INPUT	P2 CONNECTOR PINS (GROUND)
1 (S1 in pos. 13)	J10-C—+28V	u
2	J10-C—+28V	i
3	P2-T—+5V	i
4	P2-CC— -6V	DD
5	P2-FF—+10V	DD
6	P2-GG— -10V	DD
7	P2-K—+6V	M
8	P2-L—+20V	M
9	P2-AA—+15V	-i

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

S2 SWITCH POSITION	V INPUT	P2 CONNECTOR PINS (GROUND)
10	P2-t—+15V	-i
11	P2-Y—+28V	-i
12	P2-U- -15V	-i

4.3.12.3 With +28+ 0.1V applied to J10-C, -28 ± 0.1V to J10-E, and ground to P2-i, the +12V, and 115V lamps shall light. Remove above connection from System Tester after test is completed.

4.3.12.4 Connect synchro control transformer, 11CT4e to J3 of System Tester as shown in Figure I.

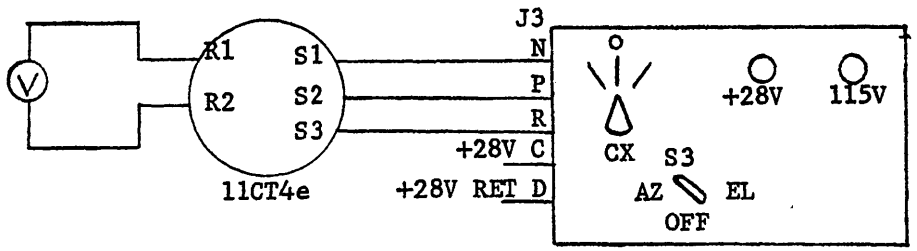


FIGURE 1

Apply +28 ± 0.1V to J3-C and gound to J3-D, the +28V and 115V lamps shall light. Place S3 to AZ position and rotate cx switch to 0 position. With voltmeter placed between R1 and R2 align dot on end of CT shaft to coincide with black line on cx body (electrical and mechanical zero). Fine adjust CT shaft until voltmeter reads 0 + 0.1 VAC. Rotate cx knob clockwise to stop, voltmeter shall read 20 ± 10% VAC. Rotate cx knob counterclockwise to stop, voltmeter shall read 20 ± 10% VAC. Place S3 to OFF position voltmeters shall read 0 volts. Remove the above connections when test is completed.

4.3.12.5 Connect the System Tester as shown in Figure II. Adjust the signal generator to apply a 1 KHz sine wave with peak-to-peak signal of 10 MV between pins A and D of connector J13. The output between pins e and K of connector P2 shall be a 1 KHz sine wave with a minimum peak-to-peak value of 15 volts. Remove the connections of Figure II when test is completed.

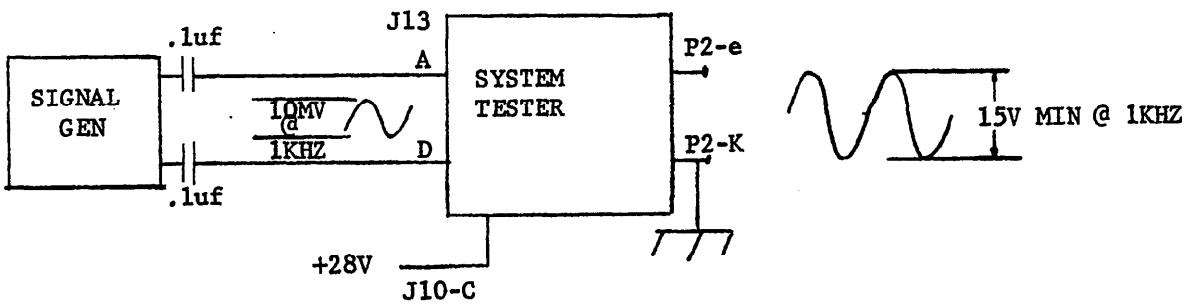


FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

4.4 Reliability Run-in. - Subsequent to the successful performance requirements of this specification the Tester shall be subjected to 50 hours of Run-in time. The Run-in shall exclude any down-time necessary for repairs. It shall include vibration, followed by at least 4 hours at -4°F and at least 4 hours at +125°F.

4.4.1 Vibration Run-in. - The vibration test shall be performed as specified in MIL-STD-810, Method 514.1, Procedure IX, Part I. Subsequent Subsequent to the vibration test, the Tester shall not have been damaged or adversely affected and shall meet the performance requirements of this specification.

4.4.2 Temperature Run-in. - The Tester shall be placed in a chamber and the temperature gradually raised at a rate not to exceed 2°F per minute until thermally stabilized at plus 125°F. Five minutes after power is applied, a complete test to the performance requirements of paragraph 4.3.12 shall commence. A second complete test to the requirements of paragraph 4.3.12 shall be accomplished during the middle of the four hour period and a third test to the performance requirements of paragraph 4,3.12 shall be accomplished during the last part of the hour test. Repeat the above tests at a temperature of -4°F. Subsequent to the above temperature extremes (plus 125°F and - 4°F) the Tester shall not have have been damaged or adversely affected and shall meet the performance requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 General. - Packaging and marking shall be in accordance with the requirements as specified in the applicable Packaging Data Sheert (PDS), 10549290.

6. NOTES

6.1 Intended Use. - The System Tester : 10549290, is one of the sub-assemblies of the XM10 which is used to evaluate simulated Target Generator Cables and Console.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Instructions for arranging of examination, test, and approval of first article (see 4.2).

6.3 Instruction for models and samples.

6.3.1 Instruction for preproduction sample. - Preproduction samples shall be as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

6.3.2 Tests. - Any change to control tests specified in 4.1.2.2 is to be so specified in the contract.

Custodian

Army - MU

Preparing Activity

Army - MU

Project No. 1240-A563

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-48419A(MU) Tester, System, 10549290

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 20261O-22T19:26Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Phila., PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:26Z
Check the source to verify that this is the current version before use.